Nathan Sauldubois

+33 6 25 96 59 80 | nathan.sauldubois@free.fr | nathansauldubois.github.io

Summary

I am a postdoctoral researcher at **New York University (NYU)**, working on probability, optimal transport, martingale optimal transport, and model risk. I obtained my **PhD in Applied Mathematics from École Polytechnique** under the supervision of **Nizar Touzi** and defended my dissertation on **November 25, 2025**. My research combines theoretical probability, optimal control, optimal transport, and distributionally robust optimization.

Skills

Mathematical Interests: Optimal transport, martingale optimal transport, probability, optimal control, mean-field games, robustness

Numerical Interests: Machine learning, deep learning for PDEs, entropic regularization, particle methods
Programming: Python, R, C++, LaTeX
Languages: French (native), English (fluent)

Publications

Published Papers

- Compoint, A., Sauldubois, N., and Touzi, N. (2026). *Sensitivity Analysis of Distributionally Robust BSDEs and RBSDEs*. *Mathematical Finance*.

Accepted Papers

- Sauldubois, N., and Touzi, N. (2026). *First-Order Martingale Model Risk and Semi-Static Hedging*. Accepted for publication in *SIAM Journal on Financial Mathematics*.

Preprints

- Sauldubois, N. (2026). *Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach*. arXiv preprint arXiv:2601.20139.
- Sauldubois, N. (2026). *Higher-Order Sensitivity Analysis of Wasserstein and Adapted Wasserstein Distributionally Robust Optimization Problems*. arXiv preprint arXiv:2609.25345.
- Sauldubois, N., and Zhang, X. (2026). *Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport*. arXiv preprint arXiv:2605.30560.

Talks

- "A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem", Séminaire Doctorant du CMAP (2023)
- "Extended L-Q Mean-Field Game with Common Noise", Hammamet "Stochastic Control and Risk" (April 2023)
- "Robust Hedging and Distributionally Robust Sensitivity", PhD Talk, Séminaire Bachelier (March 2024)
- "Model Risk and Semi-Static Hedging", XII Bachelier World Congress, Rio (July 2024)
- "Model Risk and Semi-Static Hedging", 7th Berlin Workshop on Mathematical Finance for Young Researchers, Berlin (September 2024)
- "Robust Static Hedging and Distributionally Robust Sensitivity", Hammamet Workshop, "Stochastic Control and Games for Risk and Regulation" (October 2024)
- "Adapted Wasserstein Gradient Flow and Projected Langevin Dynamics for Weak Optimal Transport", Hammamet "Stochastic Control and Risk" (April 2026)
- "Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach", XIII Bachelier World Congress, University of Bologna (July 1, 2026)
- "Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport", DMV Annual Meeting 2026, University of Konstanz (September 9, 2026)
- "Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport", 10th Eastern Conference on Mathematical Finance, Johns Hopkins University, Baltimore (October 2026)

Teaching

Courses, Exercise Sessions, and Tutorials

- **Option Pricing & Stochastic Calculus** — Graduate course, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Spring and Fall 2026)
- **C++: Financial Computing Recitations** — Graduate recitations, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Fall 2026)
- **Summer Bootcamp** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Summer 2026)
- **Winter Math Bootcamp: From Brain Teasers to Black–Scholes** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Winter 2026)
- **Probability Refreshers** — Teaching assistant for practical sessions, X–HEC Master's in Data Science for Business (2022–2025)
- **Stochastic Models in Finance** — Tutor and homework grader, École Polytechnique (2022–2025)

- **Convex Optimization and Control** — Teaching assistant for practical sessions, third-year bachelor's course (2022–2025)
- **Preparatory Classes (CPGE)** — Oral exam preparation tutor at Lycée du Parc (2019–2020)

Supervision

- Co-supervision of **Arthur Compoint**'s research internship on *continuous-time model risk for stochastic optimization problems* (2025).
- Currently co-supervising **Tianshi Liu**, who started his PhD at NYU under the supervision of Xin Zhang, on *Numerical Scheme for Projected Langevin Dynamics* (2026).

Administrative Responsibilities

- Responsible for communications and website maintenance for the **Séminaire Bachelier** (CMAP–CREST)
- Co-organizer of the **Séminaire Doctorant du CMAP**, 2023–2024
- Co-organizer, with Xin Zhang, of the invited session "**Optimal Transport in Robust Finance**" at the 2026 INFORMS Conference on Financial Engineering and FinTech

Education

École Polytechnique **2022–2025**
PhD in Applied Mathematics
Defended on November 25, 2025. Supervisor: Nizar Touzi.

ENS de Lyon **2020–2021**
M2 Advanced Mathematics

ENS de Lyon **2019–2020**
M1 Advanced Mathematics

ENS de Lyon **2018–2019**
L3 Fundamental Mathematics with a Physics Minor

Research Experience

Postdoctoral Researcher, NYU **2025–Present**

- Distributionally robust optimization in continuous time.
- Adapted Wasserstein gradient flows and applications to entropic martingale optimal transport and weak optimal transport.

CMAP, École Polytechnique **2022–2025**
PhD Research

- Mean-field games with control-dependent interactions.
- Sensitivity analysis and distributional robustness in stochastic control.
- Martingale optimal transport and model risk.

Natixis – Quantitative Research **2021–2022**
Research Internship Supervised by Pierre Henry-Labordère

- Model uncertainty in fixed-income products.
- Super-replication strategies for Bermudan swaptions with non-constant notionals.
- Quantum optimal transport methods for correlation swaps.

CMAP, École Polytechnique **2021**
Research Internship Supervised by Nizar Touzi and Hadrien De March

- Entropic optimal transport applied to healthcare policy design, in partnership with Quantev.

Oxford University **2020**
Research Internship Supervised by Julien Berestycki

- Analysis of high-dimensional branching Brownian motion and extremal particle estimates.

Université de Bordeaux **2019**
Research Internship Supervised by David Lannes

- Study of shallow-water wave equations and the behavior of solutions in bounded-depth domains.

Additional Activities

- Special Jury Prize – SMF Junior Contest (2018)